Abstraction: A Super Tool for Decisions under Uncertainty

One Mathematical Skeleton, Many Real-World Stories (C1–C4)

The journey in one line

Real-world story $\longrightarrow$ Abstract decision model $\longrightarrow$ Payoff table $\longrightarrow$ Profit/payoff calculation $\longrightarrow$ Decision criterion $\longrightarrow$ I

These are not unrelated skills. Criteria C1–C4 are successive layers of one reusable workflow.

1 A surprising connection

On Monday, Maya helps a café decide whether to install a small or large production capacity before customer demand is known. Across town, Leo advises a farmer who must plant crop A or crop B before the season turns dry or rainy. Meanwhile, Sara compares two investment projects before the market proves weak or strong.

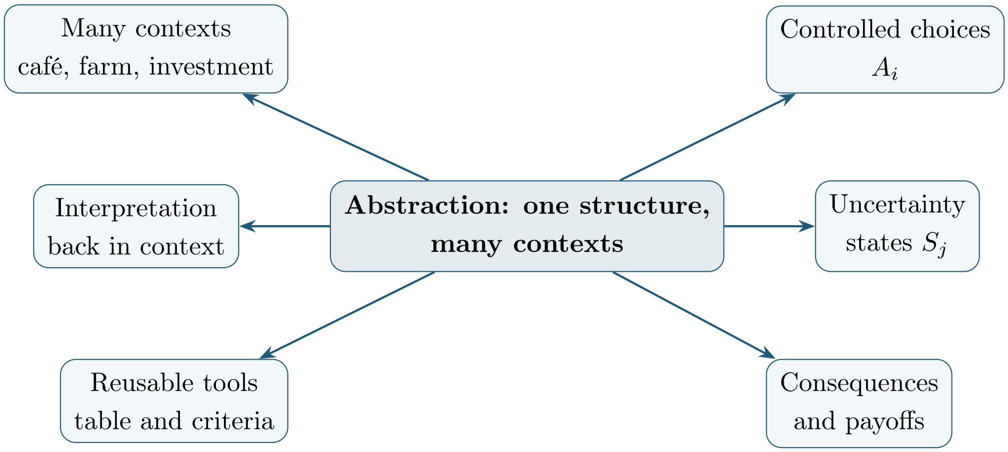
At first, the problems seem unrelated: coffee cups, crops, and investments have different prices, risks, units, and consequences. Those are the story's *clothing*. Look beneath it, however, and the same mathematical skeleton appears:

$$\boxed{\text{choice before uncertainty}} + \boxed{\text{possible states}} + \boxed{\text{payoff for each combination}}.$$

This pattern also appears when a company selects a production plan, a family evaluates a financial decision, a school enterprise plans output, or a government selects a policy. Recognising it is the super tool called **abstraction**.

2 What abstraction means

Abstraction does *not* mean ignoring reality or removing meaning. It means keeping the information that matters for the decision while temporarily setting aside distracting surface details. Once a story is translated into alternatives, states, consequences, revenues, costs, profits, and payoffs, the same tools can work in many economic and financial settings.

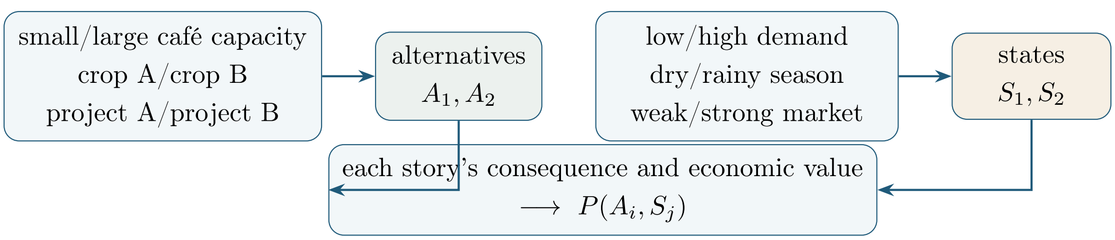


Zoom out, use the tool, zoom back in

Zoom out: replace context-specific names by A_i, S_j, and $P(A_i, S_j)$. **Apply** the general tools. **Zoom back in:** interpret the selected alternative, its risk, assumptions, and limitations in the original situation. Context tells us what the symbols mean and which data are reasonable; abstraction reveals the reusable structure.

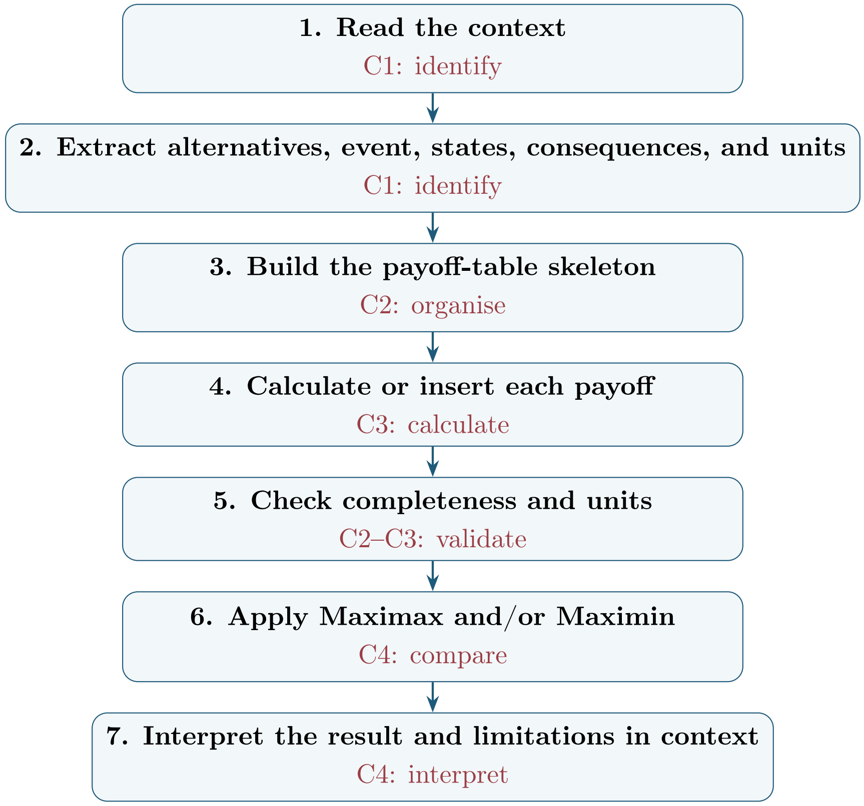
3 Different stories, the same skeleton

The nouns change, but their *roles* stay stable. In each row below, a decision-maker controls the left-hand choice, not the middle uncertain state; the combination determines the consequence and its value.



Maya's "large capacity", Leo's "crop B", and Sara's "project B" may all be labelled A_2. That does not claim they are identical in real life. It shows that each occupies the same place in a decision model.

4 The C1–C4 abstraction pipeline



5 C1: Identify the decision structure

Start by asking: Who makes the decision? What is the economic or financial context? What can that person or organisation choose? What uncertain event occurs *after* the choice? Which states can it produce? What consequence belongs to each alternative–state pair? In what units will it be measured?

Write the m alternatives and n states as

$$A_1, A_2, \ldots, A_m \qquad \text{and} \qquad S_1, S_2, \ldots, S_n.$$

The timeline prevents a common confusion:

$$\text{Choose } A_i \quad \longrightarrow \quad \text{uncertainty is resolved as } S_j \quad \longrightarrow \quad \text{consequence and payoff } P(A_i, S_j).$$

- The decision-maker controls the choice of A_i.
- The decision-maker does not control which S_j occurs.
- The combination (A_i, S_j) determines a consequence.
- The payoff assigns an economic or financial value to that consequence.

A *consequence* may first be expressed in words: for example, “the large café has unused equipment when demand is low.” Its numerical *payoff* might be a profit of -2 thousand USD. The words explain what happens; the number makes economic comparison possible. Units are part of that meaning.

6 C2: Give the structure a table

The payoff table is the bridge between a verbal story and a mathematical comparison. Repository convention places **alternatives in rows**, **states in columns**, and $P(A_i, S_j)$ in their corresponding cell:

$$\mathbf{P} = \begin{pmatrix} P(A_1, S_1) & P(A_1, S_2) & \cdots & P(A_1, S_n) \\ P(A_2, S_1) & P(A_2, S_2) & \cdots & P(A_2, S_n) \\ \vdots & \vdots & \ddots & \vdots \\ P(A_m, S_1) & P(A_m, S_2) & \cdots & P(A_m, S_n) \end{pmatrix}.$$

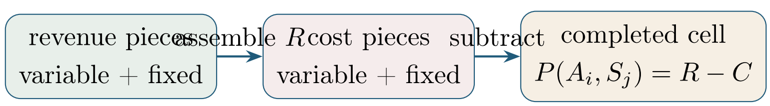
This is an $m \times n$ table: m alternative rows and n state columns. Every pair (A_i, S_j) must have exactly one cell—no gaps and no duplicates. Maya's, Leo's, and Sara's tables can therefore have the same shape even though every label and payoff has a different real-world meaning.

7 C3: Build the value in each cell

Sometimes a payoff is supplied directly and can simply be inserted. At other times, we must assemble the cell from economic information. In plain language, *revenue* is income earned, *cost* is the value of resources used, and *profit* is what remains after cost is subtracted from revenue:

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Imagine each payoff-table cell as an empty container. First assemble all its revenue pieces in $R(A_i, S_j)$. Separately assemble all its cost pieces in $C(A_i, S_j)$. Only then subtract the second container from the first.



Variable components depend on quantity; fixed components do not. A component labelled by A_i changes with the alternative, one labelled by S_j changes with the state, and $q(A_i, S_j)$ may change in every cell. A general container broad enough for many simpler stories is

$$R(A_i, S_j) = q(A_i, S_j)\big[r(A_i) + r(S_j)\big] + f_r(A_i) + f_r(S_j),$$
$$C(A_i, S_j) = q(A_i, S_j)\big[c(A_i) + c(S_j)\big] + f_c(A_i) + f_c(S_j),$$

and therefore $P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$.

Do not let the large formula hide the simple idea. Many stories use only some of its compartments:

- treat a missing component as zero;
- do not decompose a directly supplied total;
- never multiply a fixed amount by quantity;
- keep units compatible (for example, thousand USD with thousand USD);
- construct revenue and cost independently before subtracting.

For Maya, quantity might be cups sold; for Leo, tonnes harvested; for Sara, there may be no quantity at all because project profits are already supplied. The container adapts without changing the workflow.

8 C4: View one table through two decision lenses

C1–C3 construct a valid model and completed table. C4 asks how to compare its rows when no probabilities are supplied. The two criteria below answer different questions; neither is universally correct.

8.1 The optimistic lens: Maximax

Maximax asks, "What is the best outcome each alternative could achieve?" Find each row's maximum, then choose the largest of those maxima:

$$A_{\text{Maximax}} = \arg\max_i \left\{ \max_j P(A_i, S_j) \right\}.$$

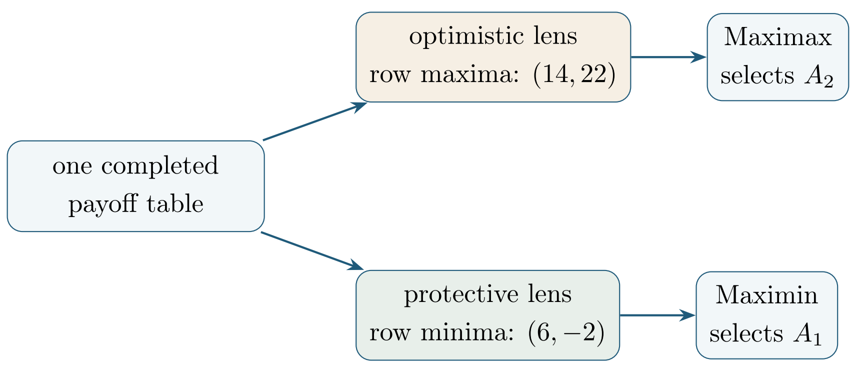
8.2 The protective lens: Maximin

Maximin asks, "What is the worst outcome each alternative could produce?" Find each row's minimum, then choose the largest of those minima:

$$A_{\text{Maximin}} = \arg\max_i \left\{ \min_j P(A_i, S_j) \right\}.$$

One small table makes the two lenses visible. Suppose Maya's café profits are in **thousand USD**:

Alternative	Low demand S_1	High demand S_2
Small capacity A_1	6	14
Large capacity A_2	-2	22



Thus, the optimistic lens favours large capacity's upside, while the protective lens favours small capacity's stronger worst case. If two alternatives share the relevant row maximum under Maximax or row minimum under Maximin, report a **tie**; do not choose one arbitrarily.

9 Zoom back into the real world

A calculation is not yet a complete recommendation. Maximax focuses on upside and may overlook severe downside. Maximin protects against the worst payoff represented in the model but may reject attractive opportunities. Neither rule uses probabilities. Maya should therefore say more than "A_2": she should name "large café capacity", state the optimistic attitude, note the -2 thousand USD downside, and explain the assumptions behind the payoffs.

Leo and Sara must do the same zoom back in. What does the selected crop imply for the farm? What risk does the project expose the investor to? Are the states complete and the monetary estimates reasonable? A model is only as useful as its alternatives, states, payoffs, units, and assumptions. A criterion is a transparent decision lens, not an automatic declaration of the one universally correct choice. Other criteria exist, but they are outside our C1–C4 journey.

10 A reusable checklist

1. What decision is being made?
2. What are the alternatives A_i?
3. What uncertain event occurs after the decision?
4. What are the states S_j?
5. What consequence belongs to each (A_i, S_j)?
6. What are the units?
7. Are payoffs supplied, or must revenue and cost be calculated?
8. Is every table cell complete?
9. Are Maximax or Maximin appropriate for the stated attitude?
10. What does the result mean in the original context?
11. What risks, assumptions, ties, or limitations must be reported?

11 Final takeaway

Keep the skeleton; restore the meaning

The context gives the decision meaning. Abstraction gives it structure. The payoff table gives it visibility. A decision criterion gives it a transparent rule for comparison. Interpretation brings the mathematics back to the real world.

The stories will keep changing. Your super tool can travel with you.